A Sharp Plaquette-Log Hessian Bound and a Finite Nonlinear Overlap Window

Abstract

We sharpen the local nonlinear $SU(2)$ block certificate by computing the exact identity Hessian of the plaquette logarithm. Transporting this exact value with the previously certified third derivative proves quotient injectivity, fiber contraction, and generated-base coercivity throughout the dependency-aware typical radius obtained from local Gaussian concentration. This closes the finite local overlap gate, while normalized density comparison and large-field charging remain open.

1 Exact identity Hessian

Let

$$F(A) = \log\bigl(e^{A_1}e^{A_2}e^{-A_3}e^{-A_4}\bigr), \qquad A_i \in \mathfrak{su}(2) \cong \mathbb{R}^3.$$

The quadratic BCH term gives a bilinear Hessian

$$D^2F(0)[A,B] = \sum_{i,j=1}^{4} C_{ij}A_i \times B_j,$$

where, up to conjugation by a diagonal sign matrix,

$$C = \begin{pmatrix} 0 & 1 & 1 & 1 \\ -1 & 0 & 1 & 1 \\ -1 & -1 & 0 & 1 \\ -1 & -1 & -1 & 0 \end{pmatrix}.$$

Theorem 1. *The bilinear operator norm of $D^2F(0)$ is*

$$m_2(0) = \|C\| = 1 + \sqrt{2}.$$

Proof. In quaternion exponential coordinates $[A,B] = 2A \times B$, so the one-half in the quadratic BCH term cancels. The singular values of C are $1+\sqrt{2}$ and $\sqrt{2}-1$, each twice. Orthogonally reduce C to two 2×2 symplectic blocks. For one block,

$$\|A_1 \times B_2 - A_2 \times B_1\| \le (\|A_1\|^2 + \|A_2\|^2)^{1/2}(\|B_1\|^2 + \|B_2\|^2)^{1/2}.$$

Combining the two blocks by Cauchy–Schwarz gives the upper bound $\|C\|$; singular vectors with orthogonal spatial directions attain it. □

If $M_3(r)$ is the certified third-derivative supremum on the joint four-link ball, the mean-value theorem yields

$$\|D^2F(A)\| \le 1 + \sqrt{2} + M_3(r)r =: m_2(r), \qquad \|DF(A) - DF(0)\| \le m_2(r)r.$$

This replaces the former global Banach-log second-derivative majorant.

2 Overlap certificate

At heat time $t = 3 \cdot 10^{-6}$, the local Gaussian theorem requires

$$r_{\mathrm{typ}} = 0.006023157807602598.$$

Substitution into the quotient Hessian and Schur-complement bounds gives:

Quantity	certified value
$m_2(r_{\mathrm{typ}})$	2.7110277092
global derivative deviation	0.0399975899
nonlinear physical singular floor	1.3742159724
total Hessian remainder	0.4746710866
fiber margin	1.5253289134
generated-base margin	0.6888073894
centered fiber transport	0.0619725685

All three local gates are strict. The old certificate stopped at radius 0.0042299674; the sharp certificate remains valid to approximately 0.0114537860, an overlap factor 1.9016 above the required radius.

3 Frontier

The theorem proves compatibility of local typical coverage and nonlinear fiber/base control at one finite regulated scale. It does not yet control the normalizing partition-function ratio after arbitrary retained boundary data, nor does it charge center, continuous, orbit, or topological complement channels. Those two estimates are the next nonperturbative target. No RG, continuum, or Hamiltonian mass-gap conclusion is asserted.